Investigating IL-6 as a Biomarker and Therapeutic Target in Schizophrenia: A Clinical Trial of Tocilizumab in Adolescents

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Abstract:

Schizophrenia is a chronic neuropsychiatric disorder that often emerges in adolescence, however diagnosis typically relies on late-stage behavioural symptoms. Increasing evidence links elevated IL-6, a pro-inflammatory cytokine, to its onset and severity. This study aims to evaluate the effectiveness of tocilizumab in reducing IL-6 levels and preventing symptom progression in adolescents at clinical high risk for schizophrenia. A 12-month, double-blind, placebo-controlled trial will be conducted with 100 adolescents (13-18 years) identified as high-risk based on symptoms, such as diabetes, obesity, and genetic predisposition. Participants will be randomly assigned to receive either tocilizumab or a placebo intravenously every four weeks for six months. IL-6 level tests and brain MRIs will be conducted monthly and biannually respectively. Preliminary expectations suggest that IL-6 levels reduced by tocilizumab will improve neuroimaging outcomes. If successful, this study would support IL-6 as a reliable biomarker in managing schizophrenia and tocilizumab as a targeted immunotherapy for early schizophrenia intervention. Future research should examine personalized treatment approaches to optimize outcomes across diverse adolescent populations.

Keywords:

Schizophrenia, neuropsychiatric disorder, interleukin 6 (IL-6), tocilizumab

Scientific Background:

Schizophrenia is a chronic and often debilitating neuropsychiatric disorder that typically emerges in late adolescence to early adulthood.[1,2] Early intervention is critical for improving long-term outcomes, yet current diagnostic methods rely on behavioral symptoms, which often appear only after significant disorder progression. Emerging evidence has highlighted a strong connection between immune dysregulation [4] — particularly elevated levels of the pro-inflammatory cytokine interleukin-6 (IL-6) and the development of schizophrenia.[5,6] Adolescents, 13-18 years, with IL-6 levels typically exceeding 5 mg/mL may represent a biologically distinct subgroup at increased risk for schizophrenia. Tocilizumab, an IL-6 receptor antagonist commonly used in autoimmune conditions, presents a promising candidate for modulating this inflammatory pathway.[3, 7, 8,] By reducing IL-6 activity, tocilizumab may alleviate symptoms and slow or prevent disease progression in high-risk adolescents. Furthermore, consistent monitoring of IL-6 levels may offer a reliable biomarker for early detection and treatment response. [10] This study seeks to evaluate the effectiveness of tocilizumab in lowering IL-6 levels and mitigating the development of symptoms in adolescents at clinical high risk for schizophrenia. Through a double-blind, placebo-controlled design, the hypothesis that IL-6 modulation can alter the course of the disorder will be tested, offering a targeted therapeutic approach and an avenue for earlier diagnosis.

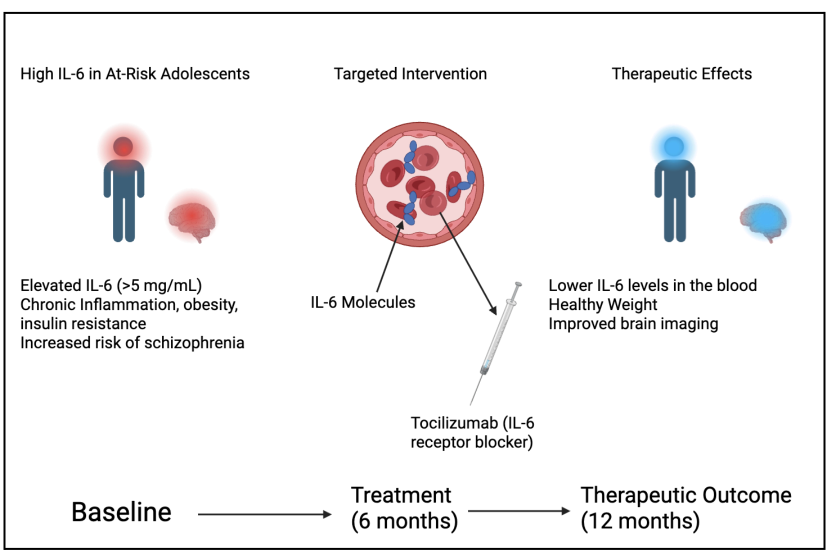

Figure 1.

Hypothesis:

If adolescents with elevated IL-6 levels are treated with tocilizumab, then their schizophrenia symptom severity and progression will be reduced compared to untreated adolescents.

Rationale:

While antipsychotic medications can reduce IL-6 levels, effects vary across different drugs, patients, symptom severity, and coexisting autoimmune conditions. [8] Limited clinical testing of tocilizumab for treatment of schizophrenia exists, however few studies target adolescents. [1,6] If IL-6 levels are measured in adolescents with high risk of schizophrenia and showing chronic inflammation, obesity, and insulin resistance, it is possible to diagnose and mitigate these symptoms earlier than the current diagnosis methods. [3] Treating schizophrenia patients using tocilizumab, then tracking IL-6 concentrations could help tailor treatment, reduce metabolic complications and offer a marker for early diagnosis. [10]

Methodology:

100 participants aged 13-18 that are clinically at high risk of schizophrenia (subtle or attenuated psychotic symptoms, noticeable functional decline in social and cognitive areas), based on prodromal symptoms and genetic disposition, will be placed in a 12 month, randomized, placebo-controlled and double-blinded study. They will undergo weekly IL-6 assessments and biannual brain imaging. Participants will be evenly divided into two groups. 50 of these

participants, the experimental group, will receive tocilizumab via intravenous infusion every 4 weeks over the course of 6 months. The other 50 participants will be the control, receiving visually matched placebo infusions containing saline at the same rate. Infusions will be administered over 60 minutes at certified outpatient clinics; pre-treatment bloodwork and symptom screening will be conducted before each dose.

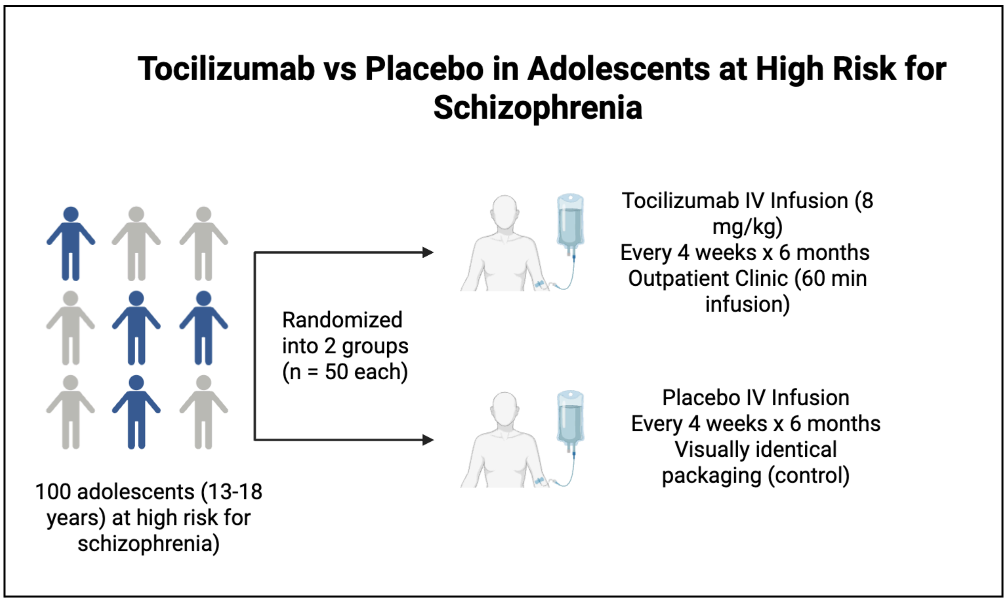

Figure 2. Study Design: tocilizumab trial in adolescents at high risk for schizophrenia

Participants will be monitored for one hour after each infusion for adverse reactions.

Monthly IL-6 levels will be conducted starting from the baseline with blood samples analyzed using therapeutic drug monitoring. [11] Results will be compared between the two groups to determine whether a sustained and statistically significant reduction in IL-6 levels over time exists in the experimental group by comparing baseline and monthly values. Brain MRIs will be conducted at the baseline and after a 6 month period to assess structural changes in the brain.

Data will be assessed with image analysis software, such as FreeSurfer or FMBRIB Software Library. [12, 13] Imaging will be conducted using a 3T MRI scanner with standardized acquisition protocols. T1-weighted structural scans will be performed to assess gray matter volume and diffusion tensor imaging will be used to evaluate white matter integrity and detect connectivity changes associated with schizophrenia. [14,15,16,17] Participants will be instructed to avoid caffeine and intense physical activity before the scan to reduce variability. Evaluations will be made on changes in grey matter volume or white volume connectivity by comparing results between the control and experimental group.

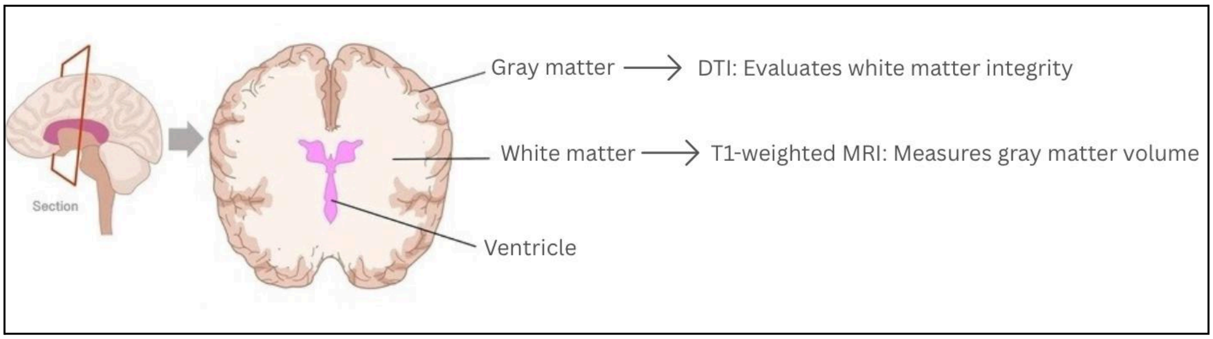

Fig. 3

Schizophrenia symptoms will be assessed monthly with several psychiatric tools including the Positive and Negative Syndrome Scale, the Brief Psychiatric Rating Scale, and the Clinical Global Impression-Schizophrenia scale. [18,19,20] The severity of symptoms over time will be analyzed with t-tests within each group, and ANOVA to compare symptom progression between the control and experimental group. [21, 22] Correlations between symptom severity and IL-6 levels will be monitored with Pearson correlation to determine if lowering inflammation is related to clinical improvement. [23]

Conclusion:

This study investigates whether tocilizumab, an IL-6 receptor blocker, can reduce schizophrenia symptoms and act as an early biomarker. If effective, differences between control and experimental groups would suggest IL-6 is linked to early onset and treatable with tocilizumab. However, treatment response may vary by demographic factors such as age, race and symptom type. Future research should explore personalized applications to ensure effectiveness across populations. These findings could support IL-6 as a biomarker and improve early diagnosis and treatment of schizophrenia.

Definition Sheet:

1. Schizophrenia = A chronic and severe neuropsychiatric disorder that affects how an individual thinks, feels and behaves. Symptoms include hallucinations, delusions, disorganized thinking and reduced emotional expression.
2. Neuropsychiatric Disorder = a condition that involves both neurological and psychiatric symptoms
3. Interleukin-6 (IL-6) = A type of pro-inflammatory cytokine (immune signaling molecule) that is often elevated in people with schizophrenia and other immune related diseases.
4. Immune dysregulation = An immune response that is directed to the body's own tissues, cells, or molecules. Can either underreact or overreact to stimuli. May manifest as autoimmune diseases where the immune system attacks healthy tissues, or as an inability to fight off infections effectively.
5. Cytokine = small proteins that help regulate the immune response. Can be proinflammatory (increase inflammation) or anti-inflammatory (reduce inflammation).
6. Tocilizumab = A medication that blocks IL-6 receptors to reduce inflammation.
7. Immune pathways = Involve the recognition of pathogens and cellular damage, leading to targeted responses like antibody production or cell-mediated immunity
8. Autoimmune conditions = When the body's natural defense system can't tell the difference between your own cells and foreign cells, causing the body to mistakenly attack normal cells.
9. Immunotherapy = Treatment that targets the immune system. In this case, to reduce brain inflammation related to schizophrenia.

10. Biomarker = A measurable substance (such as IL-6) in the body that can help diagnose or track a disease
11. Therapeutic Drug Monitoring (TDM) = Testing blood to check how much of a drug is present or how it's affecting certain markers such as IL-6.
12. FreeSurfer = An open source software suite utilized in analyzing, mapping, and visualizing neuroimaging data, particularly from MRI scans. Can process, reconstruct, and measure various aspects of the brain's structure, including cortical surfaces, subcortical structures, and white matter pathways.
13. FMBRIB Software Library (FSL) = A software package used for analyzing MRI-based neuroimaging data.
14. T1-weighted structural scan = A type of MRI that shows detailed brain anatomy, especially gray matter.
15. Gray matter = A major component of the central nervous system enabling individuals to control movement, memory, and emotions
16. White matter = A network of nerve fibers allowing the exchange of information and communication with grey matter to different parts of the brain
17. Diffusion Tensor Imaging (DTI) = An MRI method that shows the health of white matter.
18. Positive and Negative Syndrome Scale (PANSS) = A tool that measures schizophrenia symptom severity.
19. Brief Psychiatric Rating Scale (BPRS) = A scale that assesses psychiatric symptoms like depression, anxiety, hallucinations.
20. Clinical Global Impression-Schizophrenia (CGI-SCH) = Rates the severity of schizophrenia symptoms overall.

21. T-tests = A statistical tool used to determine if there's a significant difference between the means of two groups
22. Analysis of Variance (ANOVA) = A statistical method that checks if two factors are related.
23. Pearson correlation = a number between –1 and 1 that measures the strength and direction of the relationship between two variables. It assesses how well the data points fit a straight line.

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